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Climate and the general circulation (MO7021)

The Arctic Climate

March 9, 2026

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1 Introduction

The Arctic and the Antarctic are two unique regions on Earth that are characterized by extreme cold temperatures, vast ice sheets, and unique ecosystems. The Arctic is located in the Northern Hemisphere and is centred around the North Pole. In this report, we will discuss the climate in the High Arctic together with the island of Greenland to develop a deeper understanding of how this region is affected by the general circulation, climate change, and social activities in the region. This will be done using articles and books on the topic together with reanalysed data from the region.

1.1 Background - The Arctic region

When dealing with the Arctic, it is important to define the region. The Arctic is commonly defined as the area north of the Arctic Circle, which is located at 66°33" N latitude [National Snow and Ice Data Center, 2026, Serreze and Barry, 2014]. However, others define the Arctic based on the extent of the Arctic ecosystem, which requires a July mean temperature under 10°C [National Snow and Ice Data Center, 2026]. Using this definition, the Arctic can be divided into two main sub-regions, including the Arctic maritime region and the Arctic continental region according to National Snow and Ice Data Center [2026]. The important geographical locations mentioned in this report can be seen in Figure 1

The Arctic maritime region includes the Arctic Ocean and its surrounding seas such as the Bering and Greenland Seas, as well as the Labrador Sea and Baffin Bay. This region is sometimes described as a Mediterranean-type ocean due to its limited connection to the Atlantic Ocean and the Pacific Ocean [Danilov, 2000]. Looking at the bathymetry of the Arctic Ocean, one can observe that the Bering Strait, but also the Arctic Archipelago and the Barents Sea, are shallow with depths of hundreds of meters. Comparing this with the Fram Strait between Svalbard and Greenland, it is much deeper, allowing a bottleneck water flow between the Arctic and the Atlantic [Jakobsson et al., 2003]. Thus, it is very reasonable to view this ocean as a semi-isolated ocean, just like the Mediterranean.

The Arctic continental region consists of places like Svalbard, northern Canada, Russia, and the Nordic countries. However, the main landmass is Greenland, covering a large area of the Arctic. The Arctic is often split into the Low and High Arctic to distinguish between the more forested regions and the tundra up north [Bliss, 1997]. The landmass of Greenland is thus characterized by a tundra landscape, although aay from the coast it becomes more suitable to talk about a polar desert due to the lack of moisture [Charlier, 1969]. Furthermore, only 2–3% of the polar desert experiences some type of vegetation during the summer months [Bliss, 1997].

Figure 1: Arctic map showing the Arctic Circle (red dashed line) and some key locations mentioned in the text (marked as blue dots). Note that the Nordenskiöld Sea is also known as the Laptev Sea.

When describing the Greenlandic landmass, one can use the categorization developed by Köppen [1884] to categorize the terrain. Köppen divided the world's landmasses into five main climate groups: tropical (A), arid (B), temperate (C), continental (D), and polar (E). Furthermore, each subgroup could be divided into more groups depending on precipitation and later on temperature. Due to the constantly cold nature of the Greenlandic landmass, the region was only divided into two main groups: the tundra (ET) and the ice cap (EF).

The Arctic ice cap, known as the cryosphere, is an important concept for understanding the Arctic region due to its high albedo and effect on the region. On Greenland, the ice cap covers 81% of the total area [Serreze and Barry, 2014]. Arctic sea ice also plays a very important role in the region, as it turns the ocean into a land-like state. Furthermore, the melting and freezing of the ice are directly coupled with the atmospheric circulation and energy budget, as it ranges from $15 \times 10^6\,\mathrm{km}^2$ in March to $5 \times 10^6\,\mathrm{km}^2$ in September [Serreze and Barry, 2014]. The extent of the sea ice cover is determined by nearby landmasses, the inflow of warmer water through the Fram Strait, and drainage from large river systems such as the Mackenzie, Lena, Yenisey, and Ob, which are essential for the formation of sea ice [Serreze and Barry, 2014].

2 Data and Methodology

2.1 Methodology

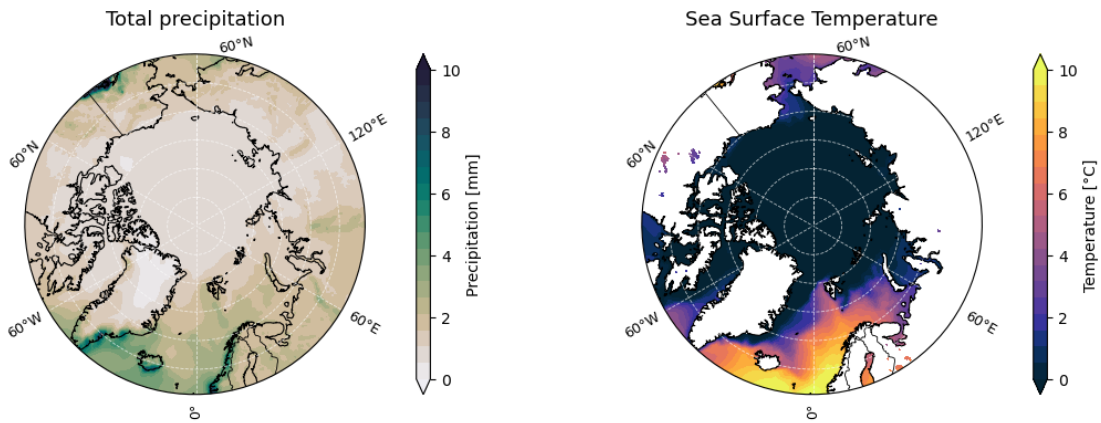
The data used in this report is based on reanalysis data, which is a combination of model simulations and observational data. The data comes from the ERA5 reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 is a global atmospheric reanalysis that provides a physically consistent estimate of the state of the atmosphere by assimilating observational data into numerical weather models [Hersbach et al., 2020]. The horizontal resolution is $0.25° \times 0.25°$ and covers the period from 2012 to 2021 by monthly means. The data were processed using Python (xarray), and graphical plots were produced using matplotlib and cartopy to visualize regional patterns and seasonal variability in the Arctic region. For more detail see Appendix A.

The reanalysis data has strengths in providing a comprehensive and continuous dataset that covers a long period of time and a wide range of variables. However, it also has limitations. The reanalysed data combine observations from various sources, including satellites, weather stations, aeroplanes, and buoys, however some regions, especially in the Arctic, have sparse observational coverage, which lead to a stronger dependence on the ones present such as satellites [Hersbach et al., 2020]. This can give rise to errors and biases such as measuring cloud cover, due to the white surface of the ice and snow leading to an overestimation of the cloud cover in the region [Hersbach et al., 2020]. Other more complex interactions, such the sea ice and atmospheric interaction, will then be heavily dependent on the simulation model's ability to capture these interactions. Thus, the uncertainties from the model simulations can be higher due to lack of observations. Another important aspect to consider is temporal resolution. The data used in this report is based on monthly mean fields, which means that it may not capture short-term variability in the Arctic region such as storms or cyclones.

2.2 The data

Above in the background section two main sub-regions of the Arctic were defined, the Arctic dessert and the Arctic tundra. It is therefore relevant to evaluate the precipitation in the region, as it is a key factor in determining the type of ecosystem present. The annual mean precipitation for the period 2012 to 2021 can be seen in Figure 2a. The figure shows that the Arctic region is generally dry, with most areas receiving less than 1 mm of precipitation per month. Higher levels of precipitation are observed in the tundra areas to the south. Thus categorizing the High Arctic as a dessert is reasonable, as it receives very little precipitation.

In Figure 2b one can observe the annual mean sea surface temperature for the period. The figure shows warmer water temperatures reaching the arctic ocean near Fram Strait, suggesting inflow of warmer water. One can also observe that the sea surface temperature is generally warmer near the Nordic Countries and the Barents Sea. However, places such as Baffin Bay, Arctic Archipelago, and the High Arctic the temperature is generally near or below freezing.

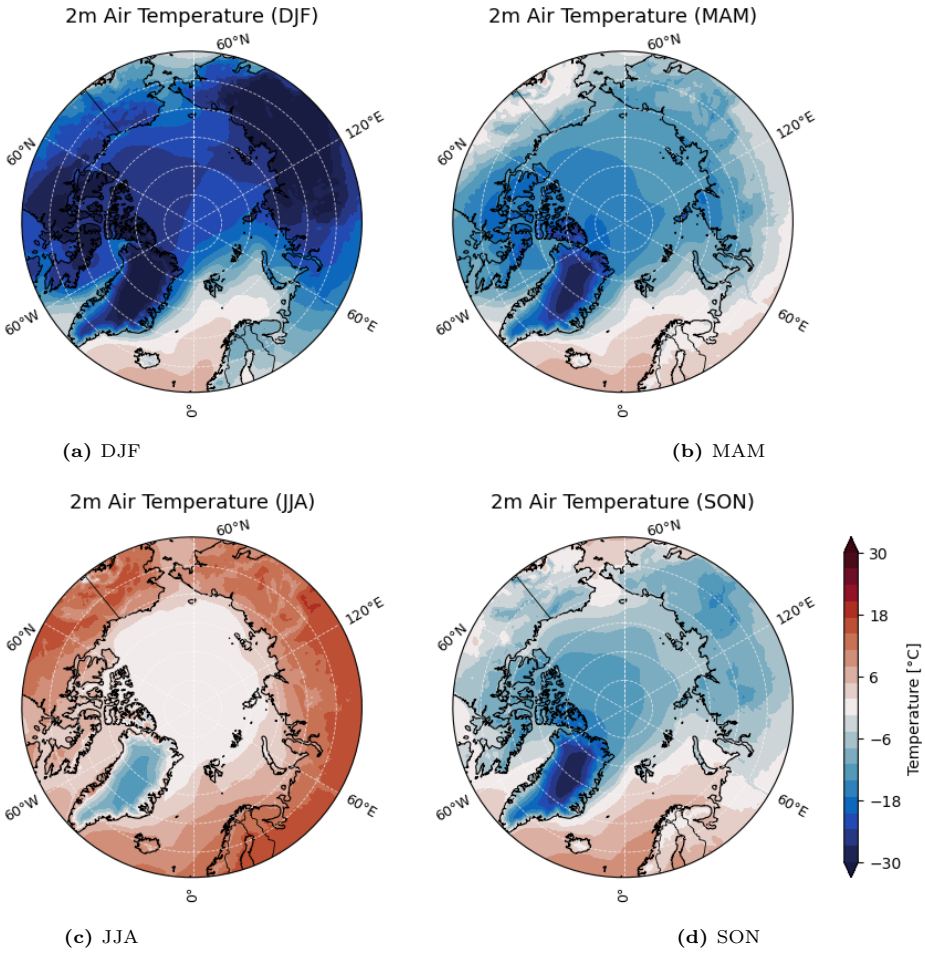


(a) Total Monthly Precipitation **(b)** Sea Surface Temperature

Figure 2: Annual monthly mean total precipitation and sea surface temperature for 2012–2021.

In Figure 2 the annual mean was calculated, since the annual precipitation and sea surface temperature were interesting to observe to evaluate the type of ecosystem present in the region. However for the data in Figure 3–5 the seasonal mean was calculated to evaluate the seasonal variability in the region. The seasonal mean was calculated by averaging the monthly mean fields for each season, where DJF corresponds to winter the months, MAM corresponds to the spring months, JJA corresponds to the summer months, and SON corresponds to the autumn months.

The air temperature change throughout the seasons can be observed in Figure 3. In the figure one can similar temperatures during the spring and the autumn, with slightly lower temperatures during the spring. The summer months are much warmer, with temperatures above freezing everywhere except for negative degrees in Greenland and zero degrees in the Arctic Ocean. The winter months are the coldest, with temperatures well below freezing across the entire region except the Norwegian Sea and parts of the Barents Sea.

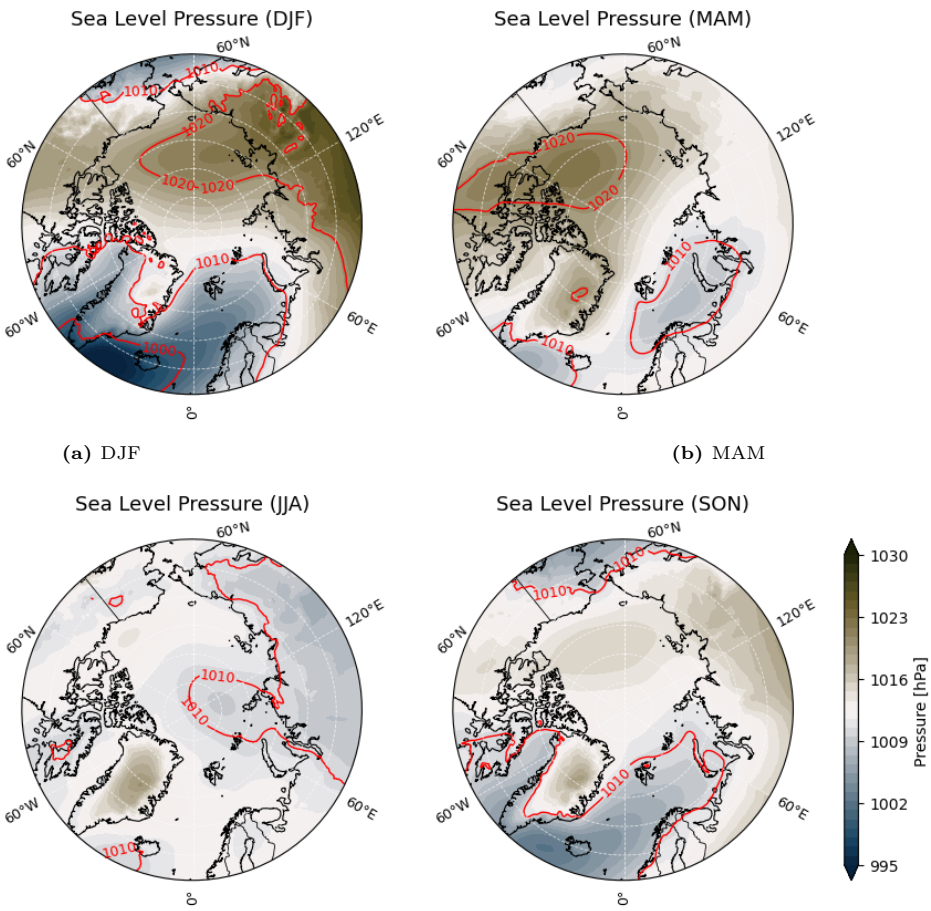


(a) DJF **(b)** MAM

(c) JJA **(d)** SON

Figure 3: Seasonal mean 2m air temperature (°C) for 2012–2021 for DJF, MAM, JJA, and SON.

The sea level pressure can be observed in Figure 4. The figure shows that the sea level pressure is generally very uniform during the summer months, with a slight high pressure system over Greenland. More interestingly is the winter months were a large pressure system can be observed over the Arctic Ocean, with a low pressure system south of the Fram Strait and south of the Bering Strait. This indicate a strong pressure gradient between the Arctic Ocean and the mid-latitudes. Altough weaker, a high pressure system can also be observed during the spring and autumn months.

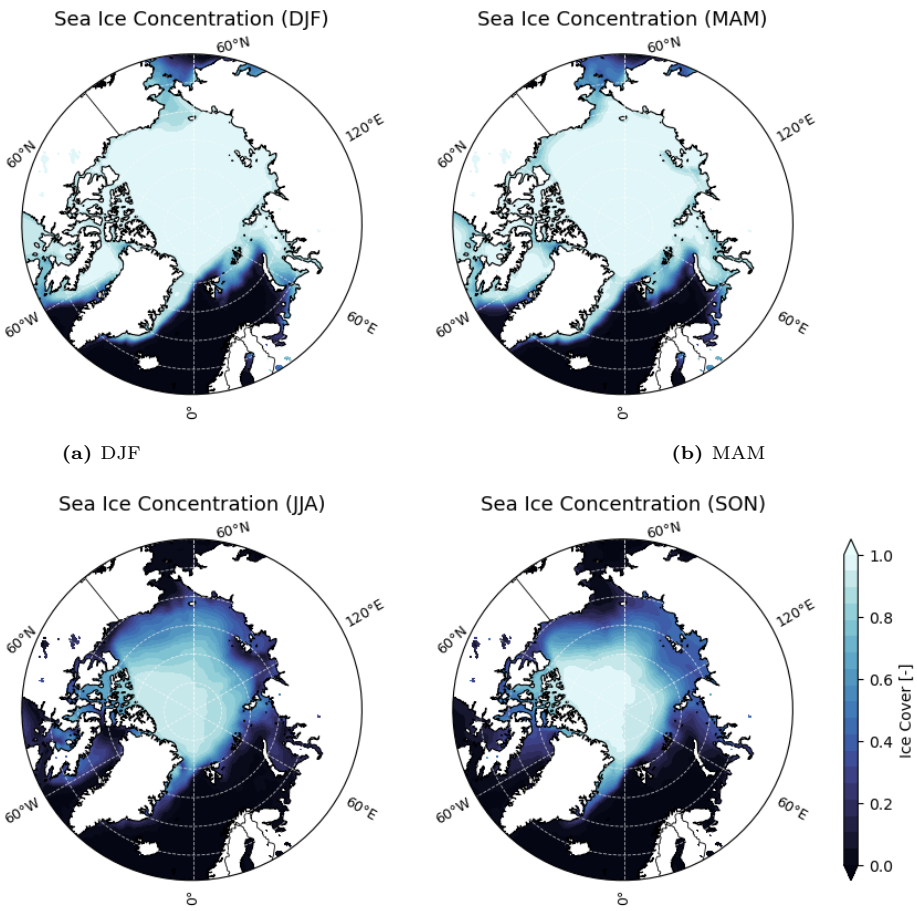


(a) DJF **(b)** MAM

(c) JJA **(d)** SON

Figure 4: Seasonal mean sea level pressure (hPa) for 2012–2021 for DJF, MAM, JJA, and SON.

The seasonal sea ice concentration can be observed in Figure 5. The figure shows that the sea ice concentration is generally very high during the winter and spring months, and much lower during the summer and autumn months. The sea ice concentration is higher in the High Arctic, but lower south of the Fram Strait and in Barents Sea. The most southern sea ice can be observed in the Bering Sea, where the sea ice concentration is around 0.5 during the winter months. More southern sea ice can be seen along the eastern coast of Greenland and in the Bering Strait.



(a) DJF **(b)** MAM

(c) JJA **(d)** SON

Figure 5: Seasonal mean sea ice concentration (-) for 2012–2021 for DJF, MAM, JJA, and SON.

3 A brief overview of the regional climate

In Figure 3–5 one could observe how the regional climate changes throughout the year. By observing the pressure one can see that extratropical cyclones are more common during the winter months, due to the large pressure gradient. One can observe that cyclones are most prominent near the Icelandic and Aleutian lows. However, according to Serreze and Barry [2014] although more cyclones occur in the storm tracks cyclones in the Arctic ocean are by no means a rare phenomena. The winter is also associated by large cryosphere extent and very cold temperatures. This can easily be explained by the polar night, leading to the only heat source being the ocean, which mostly is covered by sea ice, leading to a very cold and dry climate. Somewhat warmer temperatures can be observed during the spring, although the cryosphere extent is still large. During the summer months, the temperatures are above zero, indicating melting of the ice cap. The sea ice extent is thus at a minimum during the autumn equinox [Serreze and Barry, 2014].

4 Regional Climate and the General Circulation

4.1 The Atmosphere

An important aspect of the Arctic climate is the connection to the atmospheric circulation through the large-scale circulation patterns known as the Hadley, Ferrel, and Polar cells. The Polar cell is thus coupled with the general circulation and lifts air masses from the surface to the upper troposphere, moving them toward the North Pole, where they subside, creating a high-pressure system (as seen in Figure 4) [Hartmann, 2016]. This creates a dry desert-like climate and westerly winds, which also help drive the water southward through Ekman transport. The strength of the pressure difference between the Arctic high pressure and the mid-latitudes can be measured using the Arctic Oscillation (AO) index, which is directly linked to the North Atlantic Oscillation (NAO) [Serreze and Barry, 2014]. In Figure 4a and Figure 4c one can see that the AO index is positive during the winter, while weaker during the summer. This is because the AO index is directly linked to the polar cell, which is linked to the temperature gradient just like the Hadley cell, thus creating stronger AO during the winter.

4.2 The Ocean

The Arctic Ocean is connected to the Atlantic Ocean through the Fram Strait but is otherwise almost disconnected from the world's oceans. Thus, most oceanic water entering and exiting the Arctic Ocean passes through the Fram Strait (as seen in Figure 2b) [Serreze and Barry, 2014]. The incoming water is the Norwegian Current, which is a branch of the Gulf Stream and thus warm and saline. When this water reaches higher latitudes, it is much warmer than the atmosphere, leading to a large heat flux from the ocean to the atmosphere. Furthermore, the water becomes less saline as it mixes with freshwater from river discharge. As some water freezes, brine rejection leads to an increase in salinity and thus an increase in potential density, causing the water to sink. Interestingly, this creates an increase in temperature with depth for the initial hundred meters, which is the opposite of other oceans [Serreze and Barry, 2014].

Thus, the water that entered the Arctic Ocean through the Fram Strait eventually leaves the Arctic Ocean through the Fram Strait, but now as cold and less saline bottom water known as North Atlantic Deep Water (NADW). Additionally, as the region experiences large river discharge, more ocean water must leave the Arctic Ocean to balance the water budget. This, together with Ekman transport due to the westerlies, creates a southward transport of cold, fresh Arctic surface water through the East Greenland Current, which continues southward and contributes to the Labrador Current [Serreze and Barry, 2014]. These processes are an important part of the global meridional overturning circulation.

5 Climate Changes: Ongoing and Projected for the Future

According to Serreze and Barry [2011], the Arctic is warming at a faster rate than the rest of the world. Due to the increasing decline of sea ice during the summer months, the albedo of the region is decreasing, leading to greater absorption of solar radiation during the time when most solar radiation is available, and thus amplified heating. A study by Rantanen et al. [2022] shows that during 1979–2021 the Arctic has warmed by more than four times the global average. Thus, the Arctic is experiencing a phenomenon called Arctic amplification, which is the enhanced warming of the Arctic compared to the rest of the world. According to Intergovernmental Panel on Climate Change (IPCC) [2023], the Arctic is projected to continue warming at a faster rate than the global average, with the warming affecting ecosystems and human communities in the region.

Another important aspect of climate change in the Arctic is the thawing of permafrost [Serreze and Barry, 2014]. The thawing of permafrost is another positive feedback mechanism that can lead to further warming in the Arctic due to the release of greenhouse gases into the atmosphere, which can further enhance the greenhouse effect [Intergovernmental Panel on Climate Change (IPCC), 2023]. The thawing

of permafrost can also lead to changes in the landscape, such as the formation of thermokarst (terrain collapse caused by thawing) and thus collapse of infrastructure built on this permafrost. Furthermore, rivers in the Arctic, such as the Mackenzie, Lena, Yenisey, and Ob, may change their flow patterns due to changes in the landscape, according to Serreze and Barry [2014]. This would have significant impacts on freshwater flow into the Arctic Ocean and sea-ice formation. These changes may completely alter the heat balance of the Arctic and prevent the formation of sea ice, or even cause the Arctic Ocean to freeze to the bottom during winter. Thus, a complete change in the heat balance and important ocean currents from the region may be expected [Hartmann, 2016]. As a consequence, the Arctic is estimated to be ice-free during the summer months by the year 2050 [Hartmann, 2016]. Communities dependent on sea ice for their livelihoods, such as Indigenous peoples, may therefore face even greater challenges as sea ice continues to decline [Intergovernmental Panel on Climate Change (IPCC), 2023].

6 The Climate Impact on Societal Activities

During the last few decades, the Arctic has experienced significant changes in its climate, which have had impacts on the societal activities in the region. During the last year interests in the Arctic have increased significantly, such as the opening of new shipping routes and resource extraction due to the retreat of the ice cap [Mager, 2009]. This has led to a growing geopolitical interest for major powers in the region, such as the United States, wanting more control over the area by obtaining Greenland [Garding, 2026]. According to Garding [2026] the interests in the Arctic are due to controlling new shipping routes that passes through Greenland, as well as getting access to new minerals and oil resources in the region. One can thus speculate that the retreat of the ice cap due to climate change has led to an increased interest in the Arctic, which has led to a growing geopolitical interest in the region. One can thus speculate how this will affect the people living in the Arctic, as well as the environment in the region.

7 Conclusion

To conclude, the Arctic climate is a complex and dynamic system that is influenced by both the atmospheric circulation, the cryosphere and the ocean. Both the polar cell and the global meridional overturning circulation, have a direct impact on the Arctic climate, leading to a dry desert-like climate. Due to the forming and melting of sea ice in the cryosphere, brine rejection makes the cooling water from the Atlantic sink and return as cold bottom water. The Arctic is also experiencing a phenomenon called Arctic amplification, which is the enhanced global warming compared to the rest of the world. This has had direct impacts on the societal activities in the region, such as the opening of new shipping routes and resource extraction due to the retreat of the ice cap, making the region interesting for major powers in the region. The Arctic is thus a region with growing geopolitical interest, which may have significant impacts on the people living in the Arctic, as well as the environment in the region.

A Acknowledgements

The text in this report was spell-checked using Mistral AI's Le Chat. The data used in the report were obtained from the Copernicus Climate Change Service. The figures were created using the Python programming language along with the Matplotlib library and the Cartopy package. The code used to create the figures, alongside the report, can be found in the GitHub repository:

`https://github.com/FredrikBergelv/The-Arctic-Climate`

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